THEO GUEGAN

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Experience

Founding Engineer, Embodied AI **Feb 2026 – Present**
Opalin (Claire Delaunay, ex-VP NVIDIA Robotics) *Redwood City, CA*

- Trained efficient visuomotor policies: latent encoders over multi-view video and proprioception, deployed as real-time robot policies.
- Variance-targeted data curation: 2.8x less curated demonstrations beat unfiltered ones, lifting folding success from 20% to 98%.
- Frozen-DINOv3 phase, progress and contact estimator annotates every episode (97% top-1), cutting human labeling 70%.
- End-to-end data engine: multi-stream sync, 3D dataset viewer (multi-view, URDF) used daily for QA.
- Real-time arm control with Pinocchio IK/FK and impedance control to 0.1 mm over CAN.

Embedded AI Software Engineer **Sep 2024 – Feb 2025**
Thales Land & Air Systems *Vélizy-Villacoublay, France*

- Fully decentralized embedded AI for drone swarm coordination and control, removing the need for a permanent datalink to the ground station.
- Embedded Lua script engine and local script generation for rapid mission prototyping (Rust harness, local inference, end-to-end deployment).
- Cut new-mission deployment time by 10x and improved reliability and predictability by 4x.
- Part of Thales' COHESION demonstrator (JDEC 2024), featured in French media.

Research Assistant **Feb 2024 – Jun 2025**
CNRS Hauts-de-France, Heudiasyc Lab *Compiègne, France*

- Control architecture for a fleet of shared autonomous vehicles in dense urban environments; replaced rigid trajectory planning with waypoint-based navigation.
- Led sim-to-real deployment of the control stack onto a Renault Zoe (ROS/Python): waypoint control law, adaptive cruise control, obstacle avoidance; 99% safety in simulation.
- 1st School Award (2024), 1st place Open Category (2025), UTAC Challenge.

Business Manager **Sep 2023 – Sep 2024**
Junior UTC *Compiègne, France*

- Led technical projects for clients such as Airbus (over €30,000), coordinating student engineers.

Research

Behavior Cloning of MPC for 3-DOF Robotic Manipulators | arXiv:2606.00383

- Poster at the Workshop on RL in the Era of Imitation Learning, IEEE ICRA 2026.
- Distilled an MPC policy into neural surrogates: 3x lower inference latency at 85% success; static architectures outperform temporal ones.
- Ablation and statistical studies over traditional ML algorithms

Skills

Learning: JEPA and latent-predictive objectives, multimodal video architectures, VLAs, flow matching, imitation learning, LoRA/PEFT
Scale: multi-GPU training, efficient inference, data curation pipelines, PyTorch, JAX, Docker, CI/CD
Planning & control: MPC, Pinocchio IK/FK, RNEA, impedance control, ROS2, dora-rs, RTOS, CAN
Languages & sim: Python 3.12+, Rust, C++17, C, MATLAB, Bash; MuJoCo, Isaac Sim, Genesis, Gazebo

Education

University of Waterloo **Waterloo, Canada**
M.Sc. Artificial Intelligence, AI/ML Sep 2025 – Dec 2025

- GPA 4.0/4.0: Deep Learning (Bryan Tripp), Machine Learning (Terrence C. Stewart), Image Processing.

Université de Technologie de Compiègne **Compiègne, France**
M.Sc. Computer Engineering, Embedded & Autonomous Systems Sep 2021 – Sep 2026

- GPA 5.0/5.0: Robotics Control, Embedded Systems, Kinematics.

Projects & Awards

Open source: SO-ARM-101 and Reachy Mini manipulation stacks; autonomous drones; peillute, a distributed P2P payment system in Rust; Kaggle Brain-to-Text '25.
Hackathons: 2nd prize, Seeed x NVIDIA x Hugging Face Embodied AI (Santa Clara, 2026); 2nd prize, Open World by VLGE (San Francisco, 2026); winner, UTC x ST2I (France); Mistral AI Worldwide (San Francisco, 2026); SWARM Robotics (Canada).